SLAM Mapping Principle

1. SLAM Introduction

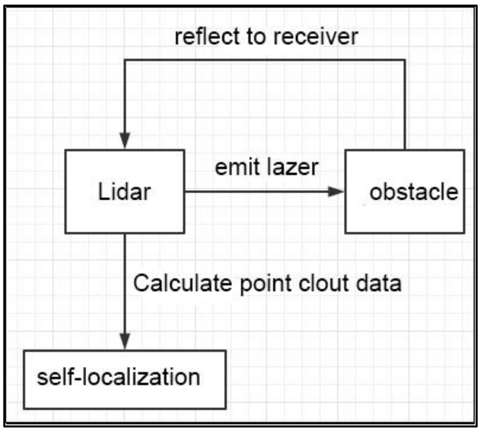
SLAM stands for Simultaneous Localization and Mapping.

Localization involves determining the pose of a robot in a coordinate system. The origin of orientation of the coordinate system can be obtained from the first keyframe, existing global maps, landmarks or GPS data.

Mapping involves creating a map of the surrounding environment perceived by the robot. The basic geometric elements of the map are points. The main purpose of the map is for localization and navigation. Navigation can be divided into guidance and control. Guidance includes global planning and local planning, while control involves controlling the robot's motion after the planning is done.

2. SLAM Mapping Principle

① Preprocessing: Optimizing the raw data from the radar point cloud, filtering out problematic data or performing filtering. Using laser as a signal source, pulses of laser emitted by the laser are directed at surrounding obstacles, causing scattering.



Some of the light waves will reflect back to the receiver of the lidar, and then, according to the principle of laser ranging, the distance from the lidar to the target point can be obtained.

Regarding point clouds: In simple terms, the surrounding environment information obtained by lidar is called a point cloud. It reflects a portion of what the 'eyes' of the robot can see in the environment where it is located. The object information collected presents a series of scattered, accurate angle, and distance information.

② Matching: Matching the point cloud data of the current local environment with the established map to find the corresponding position.

③ Map Fusion: Integrating new round data from the lidar into the original map, ultimately completing the map update.

3. Notes

1. Begin the mapping process by positioning the robot in front of a straight wall or within an enclosed box. This enhances the Lidar's capacity to capture a higher density of scanning points.

2. Initiate a 360-degree scan of the environment using the Lidar to ensure a comprehensive survey of the surroundings. This step is crucial to guarantee the accuracy and completeness of the resulting map.

3. For larger areas, it's recommended to complete a full mapping loop before focusing on scanning smaller environmental details. This approach enhances the overall efficiency and precision of the mapping process.

4. Judge Mapping Result

Finally, assess the robot's navigation process against the following criteria once the mapping is complete:

1) Ensure that the edges of obstacles within the map are distinctly defined.

2) Check for any disparities between the map and the actual environment, such as the presence of closed loops or inconsistencies.

3) Verify the absence of gray areas within the robot's motion area, indicating

areas that haven't been adequately scanned.

4) Confirm that the map doesn't incorporate obstacles that won't exist during subsequent localization.

5) Validate the map's coverage of the entire extent of the robot's motion area.